Vikram Aditya

AI & Backend Engineer · RAG Systems · LLM Fine-Tuning · Agentic AI · Azure Cloud · MLOps

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PROFESSIONAL SUMMARY

AI & Backend Engineer with 2+ years of experience across full-time and internship roles, building production AI systems (RAG pipelines, fine-tuned LLMs, agentic tool-calling) and leading development of enterprise full-stack platforms on Azure. Works across the whole lifecycle, from retrieval design and model fine-tuning to containerized cloud deployment, CI/CD and observability. Recipient of ThoughtFocus's Most Promising Newcomer award (2025).

TECHNICAL SKILLS

Generative AI & LLMs: RAG, LLM Fine-Tuning (Unsloth, QLoRA, PEFT, TRL), Hugging Face Transformers, Agentic Tool-Calling, Qwen 3, Ollama, OpenAI API, Anthropic Claude API, Prompt Engineering

RAG & Retrieval: FAISS, Sentence Transformers, Semantic Chunking, Semantic Search

Backend & APIs: Python, FastAPI, Spring Boot, Spring Security (OAuth2), Node.js, Express, Django, Flask, Ruby on Rails, REST APIs, Role-Based Access Control

Cloud & DevOps: Azure (Container Apps, Container Registry, PostgreSQL Flexible Server, Blob Storage, Front Door + WAF, Key Vault, Entra ID / Azure AD, Managed Identity, Communication Services, Application Insights), Bicep, Azure DevOps, GitHub Actions, Docker, Google Cloud Platform

MLOps & Observability: OpenTelemetry, Prometheus, Grafana, Jaeger, Redis, Rate Limiting, Circuit Breakers

ML & Data: TensorFlow, Keras, scikit-learn, NumPy, pandas, OpenCV, NLTK, Matplotlib

Frontend: React, TypeScript, JavaScript, Vite, Tailwind CSS, Streamlit

Databases: PostgreSQL (Flyway), MySQL, Redis

Languages: Python, Java, TypeScript, JavaScript, SQL, Ruby, C++

Testing & Tools: Selenium, Playwright, Faker, Git

PROFESSIONAL EXPERIENCE

Associate Engineer – AI & Backend | ThoughtFocus *Jan 2025 – Present*

Bengaluru, India · Promoted from Trainee Engineer (Feb 2026)

1. HR Knowledge Assistant: Internal Enterprise AI Assistant

Internal assistant that answers employee questions over 200+ HR policy documents and acts on IT and HR workflows.

- **RAG Pipeline:** Built retrieval with FAISS and Sentence Transformers, using semantic chunking for accurate, policy-grounded answers.
- **Fine-Tuning:** Fine-tuned a Qwen 3 model with Unsloth/QLoRA for domain-specific HR responses.
- **Agentic Tool-Calling:** Implemented tools that raise IT tickets and send automated HR notification emails from within a conversation.
- **Access Control:** Enforced role-based access control over document retrieval.

2. PitStop Connect: Talent Deployment Platform

Internal platform that runs a 60-day "PitStop" lifecycle for employees between projects, from roll-off through reservation to placement or exit. Primary developer.

- **Full-Stack Development:** Built with React + TypeScript and Java / Spring Boot on PostgreSQL: 69 REST endpoints across 16 backend modules.
- **Workflows & Scheduling:** Built reservation (block/release) and extension-approval flows, plus scheduled jobs (ShedLock, single-replica), including a nightly employee sync from the company HRMS.
- **AI Matching:** Built skill-based JD-to-candidate matching with LLM-generated hiring rationales (OpenAI gpt-4o-mini, with an Anthropic Claude fallback behind a common interface), and resume text extraction with PDFBox and Apache POI.
- **Notifications:** Built database-backed email templates editable without redeploys, an email queue, at-risk and milestone alerts and weekly digests, sent via Office 365 SMTP with an Azure Communication Services fallback.
- **Security & Integration:** Implemented Azure AD (Entra ID) SSO via OAuth2 with role-based access control, and an API-key-secured export API for app-to-app integration with an internal skills portal.
- **Cloud & CI/CD:** Provisioned Azure Container Apps, Container Registry, PostgreSQL Flexible Server, Blob Storage and Front Door with WAF rate limiting using Bicep, across dev and prod. Set up GitHub Actions CI on every PR and an OIDC-authenticated deploy workflow with versioned image tags. Added Playwright end-to-end tests.

3. ALIGN: Resource Allocation Portal

Internal portal for requesting, approving and viewing project resource allocation (Manager → PMO → Delivery Head approvals). Owned its Azure deployment and CI/CD.

- **Azure Deployment:** Set up ALIGN on Azure Container Apps with Container Registry, Key Vault and Entra ID SSO, using a user-assigned managed identity with workload identity federation, so no credentials are stored.
- **Azure DevOps CI/CD:** Designed the Azure DevOps pipeline and multi-stage Docker build: ACR cloud builds, database schema migrations guarded against data loss, health-checked Container App revisions and rollback steps. Migrated the codebase from GitHub to Azure Repos.
- **Releases & Fixes:** Ran dev releases through a scripted, health-checked deploy. Fixed SSO login in the bundled build and an HR-sync credential failure.

4. QA Automation: Adobe Experience Manager (AEM) Forms

Regression testing for enterprise AEM forms.

- **Test Automation:** Built Selenium automation with Faker-generated datasets, reducing manual regression testing effort by 60%.
- **Mentorship:** Mentored an intern on retrieval pipeline development, improving delivery quality and implementation speed.

AI/ML Engineer Intern | Tech Mahindra *Nov 2024 – Jan 2025*

Project Indus: Tech Mahindra's enterprise LLM initiative

- **RAG & Model Evaluation:** Built and evaluated RAG workflows and benchmarked 5+ model configurations, improving retrieval relevance by ∼30% using Hugging Face Transformers, PEFT, TRL and bitsandbytes on GPU-backed GCP.
- **Backend APIs:** Built and integrated Django + MySQL backend APIs for an internship application platform.

Backend Developer Intern | Mavencart *Jun 2023 – Aug 2023*

Generative AI products

- **AI Content Platforms:** Contributed to the backend of two AI products, a generative AI writer and a social media template generator, supporting content workflows that reduced manual content creation effort by ∼40%.

PROJECTS

LLM Sentinel: LLM API Gateway | *FastAPI, Redis, OpenTelemetry, Prometheus, Grafana, Docker* GitHub

- **Rate Limiting & Budgets:** Atomic per-team rate limits and token budgets enforced with Redis Lua scripts.
- **Resilience:** Circuit breaker with automatic provider fallback, verified end-to-end with fault injection.
- **Observability:** OpenTelemetry tracing (Jaeger), Prometheus metrics with alert rules, and auto-provisioned Grafana dashboards.
- **Operations:** Admin API with hot-reloaded config; full 9-container Docker Compose stack; runs at $0 via Ollama and real-wire-format OpenAI/Anthropic mocks.

Intelligent Tutoring System | *Python, FastAPI, React* GitHub

- **Adaptive Learning:** Full-stack platform that scores quizzes, recommends tiered learning resources and summarizes PDFs with PyMuPDF and NLTK; found and fixed a scoring bug that capped every quiz at 40%.

Resume Summarizer | *Python, Django, React, OpenAI* GitHub

- **LLM Document Analysis:** Extracts resume sections with GPT and generates summaries and tailored interview questions.

Face Mask Detection | *TensorFlow/Keras, MobileNetV2, Flask* GitHub

- **Computer Vision:** Trained a MobileNetV2 classifier to 97.6% validation accuracy and served it through a Flask + JavaScript web app.

More Projects | *OpenCV DNN, Stable Diffusion, Flask* GitHub

- Image colorization web app (OpenCV DNN, Flask) and a Stable Diffusion img2img ad-template generator with Pillow compositing.

EDUCATION

B.E. Artificial Intelligence and Machine Learning *2020 – 2024*

MVJ College of Engineering, Bengaluru

ACHIEVEMENTS

- **Most Promising Newcomer Award, ThoughtFocus (2025):** Presented by the Group Director & Head of HR (India) for outstanding contribution above and beyond the call of duty.